

Wei-Jin Huang

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Research Summary

I am a Ph.D. student at Sun Yat-sen University, advised by Prof. Wei-Shi Zheng, working on humanoid learning and video action understanding. My research builds physically grounded systems that learn whole-body human-humanoid interaction from human demonstrations and that understand fine-grained human actions in video. I have published as (co-)first author at CVPR, ICCV, and ECCV, and I aim to advance embodied intelligence through rigorous, impactful research.

Education

Ph.D., Computer Science and Technology – Sun Yat-sen University (SYSU), Guangzhou, China **2024.09 – Present**
Specialization: Humanoid Learning, Video Action Understanding

B.E., Network Engineering – South China University of Technology (SCUT), Guangzhou, China **2020.09 – 2024.07**
Average grade ranking 3/66

Selected Publications

- Beyond Mimicry: Learning Whole-Body Human-Humanoid Interaction from Human-Human Demonstrations**
Wei-Jin Huang, Yue-Yi Zhang, Yi-Lin Wei, Zhi-Wei Xia, Juantao Tan, Yuan-Ming Li, Zhilin Zhao, Wei-Shi Zheng
CVPR 2026 - Proceedings of the Computer Vision and Pattern Recognition Conference
Abstract: Physical human-humanoid interaction is hindered by scarce high-quality data. We leverage abundant human-human data via PAIR, a contact-preserving retargeting pipeline, and D-STAR, a hierarchical interaction policy that attains a state-of-the-art 75.4% average success rate across six interactive tasks and transfers to a real Unitree G1 humanoid.
- Modeling Multiple Normal Action Representations for Error Detection in Procedural Tasks**
Wei-Jin Huang, Yuan-Ming Li, Zhi-Wei Xia, Yu-Ming Tang, Kun-Yu Lin, Jian-Fang Hu, Wei-Shi Zheng
CVPR 2025 - Proceedings of the Computer Vision and Pattern Recognition Conference
Abstract: Error detection in procedural activities is essential for AR-assisted and robotic systems, yet prior methods rely on temporal ordering or static prototypes of normal actions. We propose AMNAR, an Adaptive Multiple Normal Action Representation framework that predicts all valid next actions and reconstructs their normal representations for comparison with the ongoing action, setting state-of-the-art across EgoPER, HoloAssist, and CaptainCook4D (+7.4% EDA / +6.5% AUC over the prior best).
- Less Static, More Private: Towards Transferable Privacy-Preserving Action Recognition by Generative Decoupled Learning**
Zhi-Wei Xia, Kun-Yu Lin, Yuan-Ming Li, Wei-Jin Huang, Xian-Tuo Tan, Wei-Shi Zheng
ICCV 2025 - Proceedings of the IEEE/CVF International Conference on Computer Vision
Abstract: Privacy-preserving action recognition aims to protect individual privacy in action videos without compromising recognition, but existing methods struggle under video domain shifts. We propose GenPriv, a transferable framework that decouples static and dynamic video features and removes privacy-sensitive content from static features, achieving strong cross-domain action recognition while preserving privacy.
- EgoExo-Fitness: Towards Egocentric and Exocentric Full-Body Action Understanding**
Yuan-Ming Li, Wei-Jin Huang (Co-First), An-Lan Wang, Ling-An Zeng, Jing-Ke Meng, Wei-Shi Zheng
ECCV 2024 - European Conference on Computer Vision
Abstract: We present EgoExo-Fitness, a 32-hour full-body action understanding dataset of 1,276 cross-view fitness sequences from 40 subjects, captured by six synchronized egocentric and exocentric cameras. It is richly annotated with two-level temporal boundaries, natural-language comments, and 1-5 action-quality scores, enabling fine-grained interpretable action analysis beyond existing datasets.

Research Interests

Humanoid Learning: embodied intelligence, whole-body control, imitation learning, human-humanoid interaction;
Video Understanding: action recognition, temporal action understanding, video representation learning

Experience

Algorithm Engineer Intern – China Telecom (Tianyi Digital Life Technology) **2023.03 – 2023.05**
• Adapted the BEiT-3 multimodal foundation model to a real-world scene-classification task, raising accuracy from 93.1% to 95.3% via vision-language fine-tuning and task-specific data curation.

Honors and Awards

- 2024** President Scholarship, Sun Yat-sen University (top 5%)
2022 Kaggle – H&M Personalized Fashion Recommendations, Silver (85/2952)
National Scholarship (1/66) • Tencent Scholarship – First Class (1/66)
2021 Kaggle – G2Net Gravitational Wave Detection, Silver (57/1219)